Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- **Lagrange Multipliers**
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Lagrange Multipliers

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The method of Lagrange multipliers method transforms the constrained problem into an unconstrained one by introducing a new set of variables called **Lagrange multipliers** λi. These multipliers are associated with the constraints $g_i(x)$. The Lagrangian function $L(x, \lambda)$ is defined as:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) = f(x) + \lambda^T g(x)$$

Here, $g(x)$ is the vector of all constraints $g_i(x)$, and λ is the vector of Lagrange multipliers λi. The key idea is to replace the hard constraint (via the indicator function) with a linear term that penalizes constraint violations in a controlled manner, making the problem easier to solve.

Lagrangian Duality

Lagrangian duality is a concept that allows us to convert the primal optimization problem (in terms of x) into a dual problem (in terms of λ). The primal problem is:

$$Minimize f(x) \quad \text{subject to } g_i(x) \leq 0 \quad \text{for all } i = 1, \ldots, m$$

This is known as the primal problem. The Lagrangian dual problem is defined as:

$$Maximize D(\lambda) \quad \text{subject to } \lambda \geq 0$$

Where:

$$D(\lambda) = \min_x L(x, \lambda)$$

This dual problem involves maximizing the function $D(\lambda)$ over the dual variables λ, while the primal problem involves minimizing $f(x)$ over x. One of the key insights from duality theory is that solving the dual problem can sometimes be easier or more insightful than solving the primal problem directly.

Minimax Inequality and Weak Duality

A fundamental concept in optimization is the minimax inequality, which states that for any function $\phi(x, y)$:

$$\max_y \min_x \phi(x, y) \leq \min_x \max_y \phi(x, y)$$

This inequality is the basis for weak duality, which asserts that the value of the dual problem is always less than or equal to the value of the primal problem. This relationship is crucial in understanding the bounds of the optimization problem and is used in proving convergence and the optimality of solutions.

Equality Constraints

In cases where the optimization problem also includes equality constraints $hj(x) = 0$ for $j = 1, \ldots, n$, these constraints can be incorporated into the Lagrangian by treating them similarly to inequality constraints but with a key difference. The Lagrange multipliers associated with equality constraints are not constrained to be non-negative, unlike those for inequality constraints.

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